Project 5 – Fragmenting wireless link

A transmitting device (TX) runs one application sending packets to one application on a receiving device (RX) via a lossy wireless link with bandwidth of C bytes per second. The TX application generates packets with size S bytes every T second, where T and S are IID random variables whose distributions are specified below. Every time a new packet is generated, the TX node:

- divides the packet into fragments of fixed size P bytes. If $S <= P$, only one fragment of size S bytes is obtained (no padding is added);
- adds a header of fixed size H bytes to each fragment;
- inserts the resulting fragments into a First-In-First-Out (FIFO) transmission queue.

The TX node transmits one fragment at a time over the wireless link, and each transmission takes F/C seconds, where F is the size (in bytes) of the fragment including the header. The TX node must wait for an acknowledgment from the receiver before starting the next transmission. In particular, the TX node retransmits the previous fragment if a negative acknowledgment is received, whereas it transmits a new fragment from the transmission queue if a positive acknowledgment is received.

The RX node considers a received fragment as corrupted with probability $p = 1 - (1 - \alpha)^F$, where α is the probability that one byte is corrupted. In that case, it sends back a negative acknowledgement otherwise it sends back a positive acknowledgement. Acknowledgments have fixed size of A bytes, hence their transmission take A/C seconds and is error-free.
The RX node can only reassemble and deliver a packet to the RX application when *all* fragments composing that packet have been correctly received.

Evaluate at least the end-to-end delay of reassembled packets for various values of P and α in the following scenario(s):

- Exponential distribution of T, uniform distribution of S.

In all cases, it is up to the team to calibrate the scenarios so that meaningful results are obtained.

Project deliverables:

a) Documentation (according to the standards set during the lectures)
b) Simulator code
c) Presentation (up to 10 slides maximum)